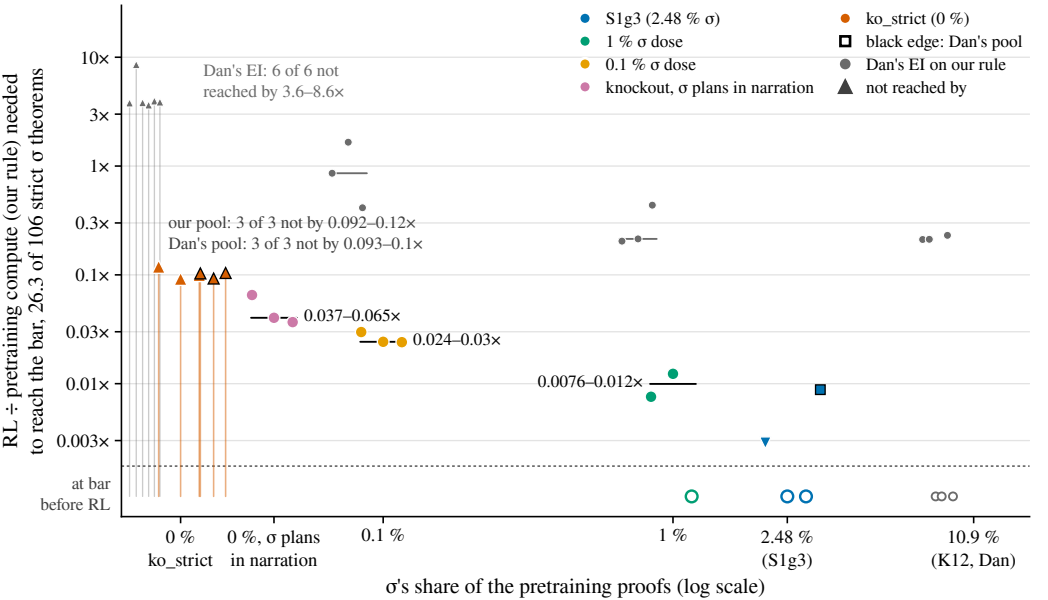

PROVISIONAL: ko_strict lines still running. One marker per seed. Bar: 26.3 of 106 held-out theorems that need classical proof by contradiction (σ, Dan's strict set), as E32 (the expected number a 32-sample draw solves), the median of our three S1g3 bases before RL. y: (6 × trained + 2 × sampled tokens) ÷ (6 × Stage-1 tokens), prompt prefill excluded. Filled: first crossing, interpolated in log y between checkpoints; ▼: at the first checkpoint (an upper bound); hollow: at the bar before RL; ▲ on a thin line: not reached by that y. Colour: GRPO with programmatic thinking traces on our pool (40–45 % of targets σ-covered); black edge: the same GRPO on Dan's 4,495 EI targets (2.7 %); grey: Dan's T1 expert-iteration ladders on this y (his own x adds re-read prefill and replay, so it is 1.5–5 × these values). Differences from Dan: GRPO, not EI; a thinking model (programmatic search traces); no pretraining replay; S1g3 bases (σ 2.48 %, not 10.9 %); doses of 0.1 % and 1 % of the proofs; ko_strict also removes the σ plans that 0.6 % of ko's records write in their narration.